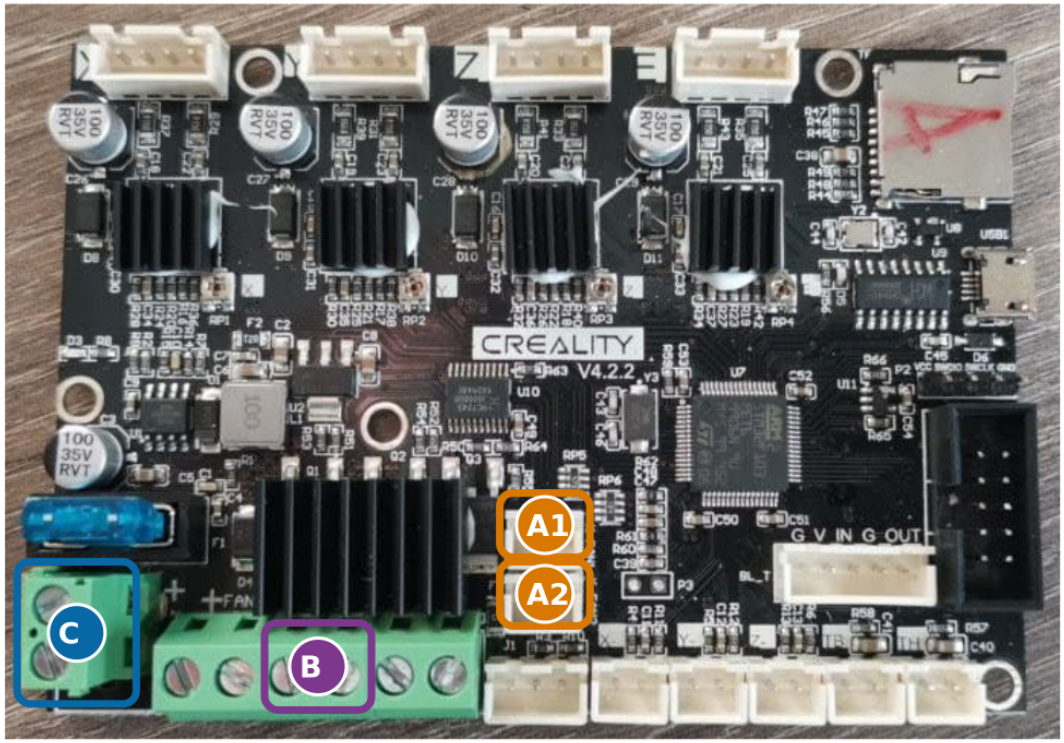
SCOPE8 — WHERE TO CONNECT THE TEST EQUIPMENT

Creality V4.2.2 / Ender-3 V2 • laser parked • heaters disconnected

Location photo below: A = controlled FAN sockets; B = BED output; C = DC input.



A — FAN TEST

K-FAN1 or K-FAN2: either socket.
Both share PA0 and the same switch.
Use ONE socket and ONE 24 V load.

B — BED TEST

Middle pair of the long green bank.
BED+ and BED−, controlled by PA2.
NOT the adjacent nozzle-heater pair.

C — SCOPE GROUND REFERENCE

Use the VERIFIED DC-input negative
or the same PSU DC-negative net.
Box C identifies the pair, NOT its − screw.

NO PHYSICAL PIN POLARITY IS GUESSED IN THIS PHOTO.

Identify + / − from verified board markings/polarity. If unclear, stop before connecting.

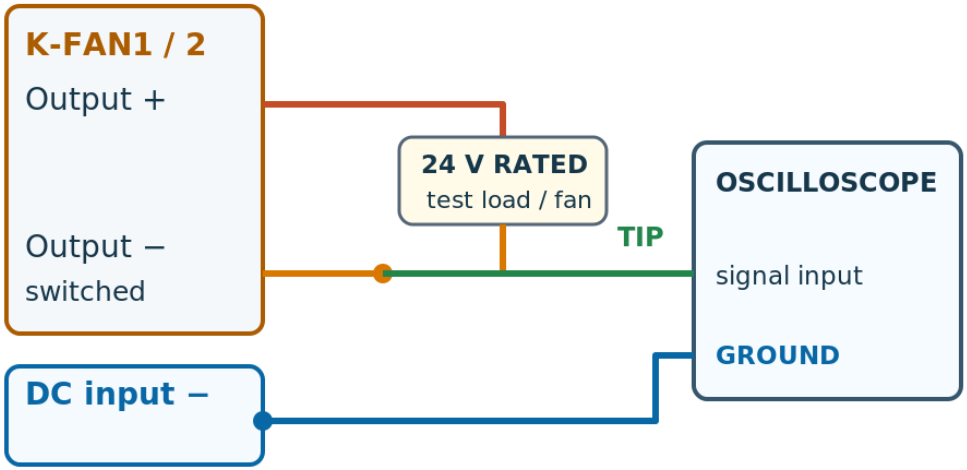
Power OFF before moving leads. Leave the working 3D Touch/probe wiring untouched.

V4.2.2 community reference photo: ender3.club/Creality_4.2.2 • not a new photo of your unit.

CONNECT ONE TARGET AT A TIME — FUNCTIONAL CIRCUITS

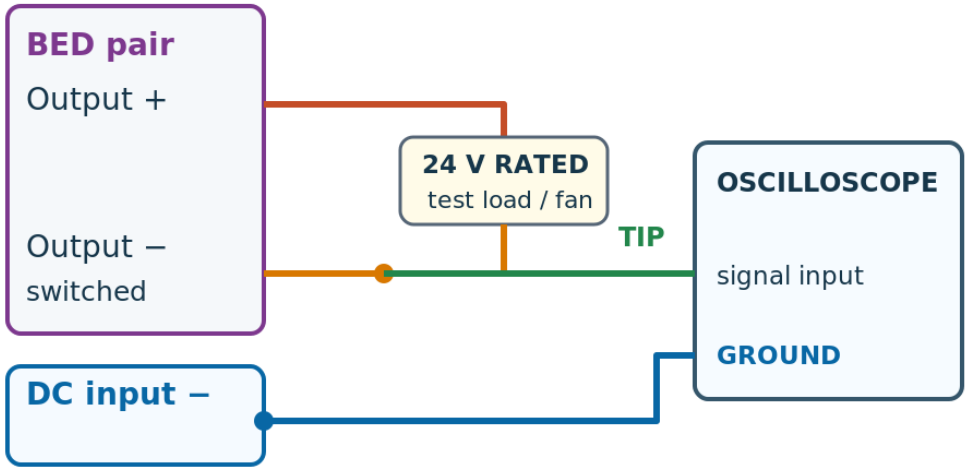
The load goes across output + / output −. Only the scope GROUND goes to board DC negative.

A: FAN test — PA0



Tip on FAN−; ground on DC input − ONLY.
PA0 drives the internal low-side switch.
Load has NO other ground/earth connection.

B: BED test — PA2



Tip on BED−; ground on DC input − ONLY.
PA2 drives the internal low-side switch.
Load has NO other ground/earth connection.

NEVER clip scope ground to FAN− or BED−: that can bypass/short the switch.

Use a suitable 24 V-rated low-current test load/fan. NO heater. NO laser. Do not defeat scope earth.

EXPECTED WITH LOAD: OFF ≈ 24 V at tip; ON ≈ 0 V at tip (relative to DC input −).

Drain waveform is inverted: more LOAD-ON duty means more time LOW at the scope tip.

SCOPE8 target: 1 kHz • Fan PA0 PWM / Bed PA2 PWM: OFF, 25%, 50%, 75%, 100%, Stop.

Use All OFF before changing target. This diagram does not operate hardware or confirm the flashed build.